Zijiang Yan

Toronto, Canada   yanzijiang97@gmail.com   (647) 870-9710   zijiang-yan   PatrickYanZ
zijiang-yan.github.io

Summary

Cloud & AI Platform Engineer with 5+ years of experience designing and operating scalable **DevOps** and cloud infrastructure across **AWS and GCP** in enterprise environments. Strong expertise in **Kubernetes**, **CI/CD automation**, and **Infrastructure-as-Code**, with a focus on cloud-native platform engineering for distributed systems. Hands-on experience building **AI/ML platforms**, data pipelines, and model deployment workflows.

Skills

Cloud & Platform Engineering: AWS, Azure, GCP (GKE), multi-cloud architecture, distributed systems
Kubernetes & Containers: Kubernetes (GKE/EKS/AKS), Docker, Helm, autoscaling, rolling deployments
CI/CD & DevOps: Jenkins, GitLab CI/CD, GitHub Actions, pipeline automation, JFrog
Infrastructure as Code: Terraform, Ansible, environment provisioning, automation
Programming: Python, Java, Docker, Bash, Groovy, JavaScript/TypeScript, SQL, C#, R
Observability & Reliability: Prometheus, Grafana, ELK, SLI/SLO, incident management

Experience

DevOps Engineer II, Scotiabank – Toronto, Canada — Apr 2026 – present

- Designed and operated scalable cloud and on-prem infrastructure supporting enterprise data platforms with 100+ integrated systems
- Built and maintained **CI/CD** pipelines and Infrastructure-as-Code (**Terraform/Ansible**) to improve deployment efficiency and consistency
- Managed Kubernetes (**GKE**) cluster and containerized workloads (**Docker, Helm**) across dev, staging, and production environments
- Implemented observability **(Prometheus, Grafana, ELK)** with **SLI/SLO**-driven monitoring for proactive incident response
- Collaborated with data engineering teams to support large-scale data ingestion, processing, and platform reliability

Senior DevOps and Cloud Engineer, Bell Media – Toronto, Canada — Oct 2021 – July 2025

- Architected and operated **Kubernetes-based CI/CD** platform supporting large-scale distributed systems
- Designed and provisioned cloud infrastructure using **Terraform**, improving deployment consistency and scalability
- Optimized CI/CD pipelines (**Jenkins/GitLab**), reducing build latency by 27% through caching and workflow redesign
- Built **Python**-based tooling for automation, diagnostics, and log analysis to reduce **MTTR**
- Implemented observability stack (**Prometheus, Grafana, ELK**) for system monitoring and alerting
- Led incident response and root cause analysis, maintaining >95% platform uptime
- Standardized artifact management and secure software delivery using **JFrog**

AI Systems & Software Engineer (Part-time), York University – Toronto, Canada Mar 2021 – present

- Designed and deployed containerized **AI/ML pipelines** supporting distributed model training
- Built scalable data processing pipelines and APIs for large-scale datasets and experimentation
- Automated research infrastructure using **Terraform and Ansible** across hybrid environments
- Developed evaluation frameworks with automated logging, telemetry, and benchmarking

Software Developer Intern, Bell Canada – Toronto, Canada May 2021 – Aug 2021

- Developed serverless applications **(AWS Lambda, Glue)** for data processing and validation
- Automated cloud-based data pipelines improving operational efficiency and observability

Education

York University, MASc in Electrical and Computer Engineering – Toronto, Canada Jan 2026 – Dec 2027

- Focus: **AI**-driven wireless communication, **LLM**-assisted decision systems, and **Quantum** Machine Learning
- Advisor: Prof. Hina Tabassum, Prof. Ping Wang

York University, BSc (Hons.) in Computer Science & Statistics – Toronto, Canada Sept 2016 – Aug 2021

- Focus on AI, cloud computing, and data-intensive applications
- Capstone: Optimizing V2I Communication for Autonomous Driving using RL-based frameworks

Publications

Semantic-Aware Adaptive Video Streaming Using Latent Diffusion Models for Wireless Networks July 2025
Zijiang Yan*, Jianhua Pei*, Hongda Wu, Hina Tabassum, Ping Wang
10.1109/MWC.001.2500068 (IEEE Wireless Communications)

Generalized Multi-Objective Reinforcement Learning with Envelope Updates in uRLLC-enabled Vehicular Networks June 2025
Zijiang Yan, Hina Tabassum
10.1109/TVT.2025.3580502 (IEEE Transactions on Vehicular Technology)

Hierarchical and Collaborative LLM-Based Control for Multi-UAV Motion and Communication July 2025
Zijiang Yan*, Hao Zhou*, Jianhua Pei, Hina Tabassum
10.48550/arXiv.2506.06532 (ML4Wireless Workshop @ ICML 2025)

Hybrid LLM-DDQN based Joint Optimization of V2I Communication and Autonomous Driving Feb 2025
Zijiang Yan, Hao Zhou, Hina Tabassum, Xue Liu
10.1109/LWC.2025.3539638 (IEEE Wireless Communications Letters)

CVaR-Based Variational Quantum Optimization for User Association in Handoff-Aware Vehicular Networks June 2025
Zijiang Yan, Hao Zhou, Jianhua Pei, Aryan Kaushik, Hina Tabassum, Ping Wang
10.1109/ICC52391.2025.11161596 (IEEE ICC 2025)

Optimizing Vehicular Networks with Variational Quantum Circuits-based Reinforcement Learning May 2024
Zijiang Yan, Ramsundar Tanikella, Hina Tabassum
10.1109/INFOCOMWKSHPS61880.2024.10620888 (IEEE INFOCOM 2024 Poster)

Multi-UAV Speed Control with Collision Avoidance and Handover-Aware Cell Association - DRL with Action Branching Dec 2023
Zijiang Yan, Wael Jaafar, Bassant Selim, Hina Tabassum
10.1109/GLOBECOM54140.2023.10436730 (IEEE GLOBECOM 2023)

Reinforcement Learning for Joint V2I Network Selection and Autonomous Driving Policies Dec 2022
Zijiang Yan, Hina Tabassum
10.1109/GLOBECOM48099.2022.10001396 (IEEE GLOBECOM 2022)

Concept Drift-Aware Load Forecasting for IoT Energy Systems Leveraging Times-Net-Based Two-Layer Framework Mar 2026
Yixiang Huang, ***Zijiang Yan***, Jianhua Pei, Jinfu Chen
(Under Review)

EMFusion: Conditional Diffusion Framework for Trustworthy Frequency Selective EMF Forecasting in Wireless Networks Dec 2025
Zijiang Yan*, Yixiang Huang*, Jianhua Pei, Hina Tabassum, Luca Chiaraviglio
10.48550/arXiv.2512.15067 (Under Review)

Professional Activities

Conference Session Chair: IEEE ICC 2025, IEEE GLOBECOM 2023

Technical Program Reviewer: NeurIPS, IEEE ICC, IEEE GLOBECOM, IEEE WCNC, IEEE VTC, IEEE DySPAN

Journal Reviewer: IEEE TMC, IEEE TWC, IEEE TCOM, IEEE TNSE, IEEE COMML, IEEE TNSM, IEEE WCM, IEEE IOTM

Honors and Awards

- 3rd Place – Student Innovation Competition on Sustainable Space Communications, 2025
- Bell Geekfest Inspiring Lightning Talk, 2025
- Silver Award – China Youth Creative Competition (Technology Innovation Track), 2024
- Bell Geekfest Best Presentation, 2023
- Lassonde Undergraduate Research Award (LURA), York University, 2022
- Dean's List, Lassonde School of Engineering, 2018